



Region-aware graph neural networks for real-time drying simulations of unsaturated porous media - Food drying digital twin framework

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ARTICLE INFO

Keywords:

Graph neural network
Food drying
Computational fluid dynamics (CFD)
Digital twin
Scientific machine learning

ABSTRACT

Real-time simulation and the integration of digital twins necessitate the development of computationally efficient surrogate models for contemporary food drying systems, in which complex physical processes must interact dynamically with virtual representations. Traditional modeling approaches for food drying are subject to notable limitations: empirical models often lack physical fidelity, whereas high-fidelity computational fluid dynamics (CFD) simulations require the resolution of tightly coupled heat and mass transfer processes among vapor, liquid water, and air phases within unsaturated porous media, rendering them computationally prohibitive for real-time applications. This work proposes a region-aware graph neural network (GNN) surrogate model for food drying simulation by leveraging mesh-based and region-specific representations of the drying process. The proposed GNN architecture, inspired by MeshGraphNet with specialized masked message passing, handles the heterogeneous domains and sharp discontinuities characteristic of coupled heat and mass transfer in food drying. The GNN model was trained on a limited dataset comprising 28 CFD simulations of hot-air drying of a potato slice under varying ambient conditions, generated using a CFD framework validated against numerical results from the literature and laboratory-scale experimental data. Despite this small training set, the model delivers predictive performance for operating conditions beyond those seen during training. Single-step predictions achieved RMSE values of 0.17 °C for temperature (T), 0.26% for relative humidity (RH), and 0.0028 for liquid saturation (S_w). Comparative analysis with standard message passing revealed a threefold reduction in interfacial absolute error (0.003 vs. 0.008) and elimination of nonphysical field propagation across domain boundaries. The model maintained performance on the training domain edges, with moisture ratio curves achieving $R^2 > 0.97$ for 10-step rollouts ($R^2 = 0.989$ interior domain; $R^2 = 0.979$ edge domain) while preserving the characteristic S-shaped drying curves in longer autonomous simulations ($R^2 = 0.784$ and 0.873 for full rollouts of the interior and edge-of-domain test cases, respectively). This framework is applicable to real-time process control, design optimization, and digital twin implementations in food manufacturing.

1. Introduction

The rapid digitalization of manufacturing and the rise of smart processing technologies are steering food production toward data-driven, energy-efficient, and highly controlled operations. In this context, preservation technologies must balance product quality, resource efficiency, and sustainability. Drying remains a fundamental preservation process, as moisture removal inhibits microbial growth, slows biochemical degradation, and enhances storage stability (Alp and Bulantekin, 2021). Its significance is reflected in the scale of global food

losses: nearly one-third of food produced for human consumption is lost or wasted annually, with inadequate preservation and post-harvest processing among the primary causes (Gustavsson et al., 2011; Khan et al., 2022). Meanwhile, industrial drying is highly energy-intensive, accounting for approximately 12–25% of industrial energy consumption in developed countries (Gustavsson et al., 2011; Karim and Hawlader, 2005). These sustainability and energy constraints underscore the need for intelligent, model-driven drying systems capable of enhanced monitoring, predictive optimization, and real-time control (Khan et al., 2022; Lee et al., 2020; Muñoz López et al., 2024).

This article is part of a special issue entitled: Digital Twins published in Journal of Food Engineering.

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<https://doi.org/10.1016/j.jfoodeng.2026.113041>

Received 28 June 2025; Received in revised form 19 February 2026; Accepted 20 February 2026

Available online 24 February 2026

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To improve monitoring, control, and prediction of drying kinetics, a wide spectrum of mathematical models has been developed, ranging from empirical correlations to detailed mechanistic simulations. Empirical and semi-empirical models are typically fitted to experimental drying data (Dadali and Özbek, 2008; Overhults et al., 1973; Page, 1949; Henderson and Pabis, 1961). While computationally efficient and widely adopted, these models lack physical realism because they assume uniform moisture distribution and neglect internal spatial gradients. As a result, small parameter uncertainties can propagate into significant errors in predicted temperature and moisture evolution (Curcio et al., 2008; Irudayaraj et al., 1992).

Empirical modeling has also evolved to include deep learning, where feed-forward artificial neural networks (ANNs) are used as black-box regression models to predict drying behavior with low prediction errors (Liu et al., 2007; Momenzadeh et al., 2011). Recurrent neural networks (RNNs) and long short-term memory (LSTM) models capture temporal dependencies in sequential data and are well suited for modeling time-dependent moisture and temperature dynamics during drying (Simonić et al., 2025). Meanwhile, convolutional neural networks (CNNs) have been applied to visual inspection and moisture estimation by extracting spatial features from image data (You et al., 2024).

Despite these advances, neural networks used within empirical and purely data-driven modeling frameworks exhibit fundamental limitations when applied to physical drying processes. Standard ANNs treat drying as a fixed input–output mapping, limiting their ability to incorporate new variables or geometry changes (Zhao et al., 2024). RNNs capture temporal dynamics but neglect spatial relationships, while CNNs assume structured grids and therefore struggle with irregular geometries and moving boundaries (Pfaff et al., 2021; Zhao et al., 2024). More broadly, these models lack embedded physical constraints and often produce unreliable or physically inconsistent predictions outside their training domain, reducing their suitability for digital twin and decision-support applications (Vasiliauskaite and Antulov-Fantulin, 2024).

In contrast, computational fluid dynamics (CFD) models rigorously solve coupled heat and mass transfer equations to provide high-fidelity spatiotemporal predictions of temperature and moisture fields (Datta, 2007a). By representing food as a porous medium, these models capture internal gradients, evaporation fronts, and moisture heterogeneity, enabling virtual drying experiments (Mondal and Datta, 2010). However, their substantial computational demands—often involving millions of state variables—limit their practicality for design, real-time control, digital twin deployment, and iterative process optimization (Lepsien and Schaum, 2025). This gap has motivated the development of modeling strategies that balance accuracy with computational efficiency.

Reduced-order models (ROMs) offer a promising alternative by capturing essential system dynamics in a low-dimensional form. Projection-based techniques such as Proper Orthogonal Decomposition (POD) and Dynamic Mode Decomposition (DMD) derive dominant modes from high-fidelity data and evolve a small set of coefficients instead of solving full-field equations (Lepsien and Schaum, 2025). These approaches enable real-time surrogate simulations and have been used to reproduce overall drying trends much faster than CFD (Ghosh and Datta, 2023).

However, classical ROMs struggle with the varying geometries and mesh structures typical of food drying. They generally assume consistent spatial discretization on fixed or structured grids. When applied to unstructured or parametric meshes, they often rely on mesh mapping or interpolation, which can introduce errors and reduce geometric fidelity and physical conservation (Fresca and Manzoni, 2022; Pichi et al., 2024).

Graph neural networks (GNNs) have recently emerged as a powerful data-driven alternative for complex multiphysics systems (Vasiliauskaite and Antulov-Fantulin, 2024; Zhao et al., 2024).

Operating on graph-structured data, GNNs naturally handle irregular meshes and heterogeneous domains by representing spatial elements as nodes and interactions as edges while preserving moving boundaries and multiphysics coupling (Wu et al., 2021). Their permutation invariance provides robustness to mesh point numbering, topology changes, and shape variations, enabling deployment across different products and operating conditions (Xu et al., 2025). Through spatiotemporal message passing, GNNs capture both local heat and mass transfer and global system behavior, achieving near-solver-level accuracy with one to two orders of magnitude faster predictions (Choi and Kumar, 2024; Pfaff et al., 2021). Consequently, GNNs provide a promising framework for modeling moisture and temperature fields, moving phase boundaries, and evolving morphologies, supporting efficient and generalizable digital twin and decision-support applications.

The objective of this study is to develop a region-aware graph neural network surrogate model for multiphysics drying of porous media that explicitly accounts for distinct physical domains (food and hot air) to accurately capture coupled field interactions. A MeshGraphNet-based framework is developed using graph-structured mesh representations and domain-specific node features to distinguish physical regions. The methodology is implemented and validated on a two-dimensional potato-slice drying benchmark to evaluate spatial accuracy, temporal rollout performance, and computational efficiency. The resulting framework enables physics-consistent and computationally efficient digital twin workflows for industrial food drying applications.

2. Methodology

2.1. Governing equations

The simulations generate synthetic data using mechanistic heat and mass transfer equations for porous media (Datta, 2007a). The porous medium is assumed to be hygroscopic, unsaturated, and rigid, consisting of three phases: solid (s), liquid (l), and gas (g). The liquid phase is pure water (w), including both free and bound water. The gas phase combines dry air (a) and water vapor (v). It is important to address the primary assumptions employed in food drying models described in (Purlis, 2019), which include.

- The medium exhibits continuity and homogeneity, as defined by a representative elementary volume (REV).
- Local thermal equilibrium exists within the multiphase system.
- The liquid phase maintains incompressibility.
- Evaporation and condensation processes occur locally within the material between liquid water and water vapor.
- The gas phase behaves as an ideal mixture of perfect gases.
- Chemical reactions are absent from the system.
- Gravitational effects are considered negligible.

The transient mass conservation equations for water, vapor, and air, respectively, are as following (Datta, 2007a):

$$\frac{\partial c_w}{\partial t} + \nabla \cdot (\vec{n}_w) = -i_{\text{evap}} \quad (1)$$

$$\frac{\partial c_v}{\partial t} + \nabla \cdot (\vec{n}_v) = i_{\text{evap}} \quad (2)$$

$$\frac{\partial c_a}{\partial t} + \nabla \cdot (\vec{n}_a) = 0 \quad (3)$$

where $c_w = \rho_w \varepsilon S_w$, $c_v = \rho_v \varepsilon S_g$, and $c_a = \rho_a \varepsilon S_g$ are the water, vapor, and air mass concentrations. In these definitions, ρ is the density of each corresponding phase, ε is the solid matrix porosity, S_w and S_g are the water and gas saturations in the pore volumes, respectively. Also, $\vec{n}_w$, $\vec{n}_v$, $\vec{n}_a$ represent the convective mass flux for each of the water, vapor,

and gas phases. By considering all assumptions discussed previously, the Darcy's law for each phase reads:

$$\vec{n}_w = -\rho_w \frac{k_{rw} k_\varepsilon}{\mu_w} \nabla p_g - D_w \rho_w \varepsilon \nabla S_w \quad (4)$$

$$\vec{n}_v = -\rho_v \frac{k_{rg} k_\varepsilon}{\mu_g} \nabla p_g - \rho_g \varepsilon S_g D_{eff,g} \nabla \left(\frac{\rho_v}{\rho_g} \right) \quad (5)$$

$$\vec{n}_a = -\rho_a \frac{k_{rg} k_\varepsilon}{\mu_g} \nabla p_g + \rho_g \varepsilon S_g D_{eff,g} \nabla \left(\frac{\rho_v}{\rho_g} \right) \quad (6)$$

where ρ_g , μ_g , and p_g are the density, dynamic viscosity, and total pressure of the gas phase (air + vapor). Moreover, k_ε is the intrinsic or porous matrix permeability where k_{rw} and k_{rg} are the relative permeabilities for water and gas phases defined below (Purlis, 2019),

$$k_{rw} = \begin{cases} \left(\frac{S_w - S_{wi}}{1 - S_{wi}} \right)^3, & S_w > S_{wi} \\ 0, & S_w \leq S_{wi} \end{cases} \quad (7)$$

$$k_{rg} = \begin{cases} 1 - 1.1 S_w, & S_w < 1/1.1 \\ 0, & S_w \geq S_{wi} \end{cases} \quad (8)$$

In Eq. (7), S_{wi} is the irreducible liquid phase saturation, describing the saturation of the liquid phase that will remain inside the porous medium. It is worth noting $D_{eff,g}$ is the molecular diffusion of the gas phase follows the Millington and Quirk equation (COMSOL, 2025) described in Eq. (9) with the binary vapor-air diffusivity $D_{va} = 2.6 \times 10^{-5} \text{ m}^2/\text{s}$ at $T = 30^\circ \text{C}$ (Bolz, 2019). Finally, D_w is known as the capillary diffusion coefficient (Datta, 2007b; Purlis, 2019) shown in Eq. (10) with ρ_s defined as porous material density.

$$D_{eff,g} = D_{va} \varepsilon \left(\frac{4}{3} \right) S_g \left(\frac{10}{3} \right) \quad (9)$$

$$D_w = 1 \times 10^{-8} \exp \left(-2.8 + 2 \times \left(\frac{\varepsilon S_w \rho_w}{(1 - \varepsilon) \rho_s} \right) \right) \quad (10)$$

The energy equation closure (Datta, 2007a) in the porous medium is given by Eq. (11) and Eqs. (12) and (13) describe the effective thermophysical properties.

$$(\rho c_p)_{eff} \frac{\partial T}{\partial t} + \nabla \cdot \left(\vec{n}_v c_{p,v} \Delta T + \vec{n}_a c_{p,a} \Delta T + \vec{n}_w c_{p,w} \Delta T \right) = \nabla \cdot (k_{eff} \nabla T) - \lambda \dot{I} \quad (11)$$

$$(\rho c_p)_{eff} = \rho_s c_{p,s} (1 - \varepsilon) + \rho_w c_{p,w} \varepsilon S_w + \rho_g c_{p,g} \varepsilon (1 - S_w) \quad (12)$$

$$k_{eff} = k_s (1 - \varepsilon) + k_w \varepsilon S_w + k_g \varepsilon (1 - S_w) \quad (13)$$

In Eq. (11), $c_p \Delta T$ represents the enthalpy of each phase, $\lambda = 2.435 \times 10^6 \text{ J/kg}$ is the latent heat of evaporation and, $\dot{I}$ is the spatial distribution of evaporation rate. The latent heat of evaporation used in this study corresponds to that of free water at approximately room temperature. It is well established that the effective latent heat of vaporization of water in food materials increases as moisture content decreases due to stronger water-solid interactions, and that it varies with both moisture content and temperature ((Mujumdar, 2014; Rahman, 2009)). In the later stages of drying, the remaining water is largely bound or weakly bound to the solid matrix rather than present as bulk free water ((Lewicki, 2004; Singh and Heldman, 2013)). Nevertheless, for weakly bound water, the effective latent heat is often only moderately higher than that of free water, and many drying analyses adopt the latent heat of free water as a practical approximation ((Maroulis and Sarabakos, 2003; Mujumdar, 2014)). The assumption is reasonable within the scope of the present work, while future studies could improve accuracy at low moisture

contents by incorporating moisture- and temperature-dependent latent heat values and targeted experimental validation.

2.2. GNN architecture and message passing

Graph neural networks (GNNs) provide a framework for learning from data that is inherently relational and geometric. In the context of mesh-based simulations, nodes on the computational mesh correspond to points in the spatial domain, and edges reflect adjacency or connectivity within that mesh. Each node carries features such as temperature, moisture content, or other physically relevant quantities, while edges encode geometric or topological information, including the relative position of nodes and shared boundaries. In this study, we draw inspiration from DeepMind's MeshGraphNet framework (Pfaff et al., 2021), which addresses mesh-based physics simulations through a graph-centric methodology that systematically encodes, processes, and decodes physical states via a learned model. The framework operates on a graph $G = (V, \mathcal{E})$ where V represents vertices (nodes) and $\mathcal{E}$ represents edges. Each node $v_i \in V$ contains features representing physical quantities (e.g., temperature, velocity), while each edge $e_{ij} \in \mathcal{E}$ connecting nodes v_i and v_j contains relational information such as relative positions and distances. Mathematically, the overall architecture transforms input features $\mathbf{X}^{(0)}$ to output predictions $\mathbf{Y}$ through sequential stages:

$$\mathbf{Y} = f_{\text{decoder}} \left(f_{\text{processor}} \left(f_{\text{encoder}} \left(\mathbf{X}^{(0)} \right) \right) \right) \quad (14)$$

The MeshGraphNet decomposes the network into three conceptual components.

a) **Encoder:** A function ϕ^{encoder} that maps the raw node $\mathbf{x}_i^{(0)}$ and edge features $\mathbf{e}_{ij}^{(0)}$ into latent representations:

$$\mathbf{h}_i^{(0)} = \phi_{\text{node}}^{\text{encoder}} \left(\mathbf{x}_i^{(0)} \right) \quad (15)$$

$$\mathbf{h}_{ij}^{(0)} = \phi_{\text{edge}}^{\text{encoder}} \left(\mathbf{e}_{ij}^{(0)} \right) \quad (16)$$

These encoding functions transform the raw temperature, humidity, and liquid saturation measurements at each mesh point into a unified mathematical representation that the network can process to learn the underlying physics.

b) **Processor:** In the physical mesh, each processing layer simulates one step of heat and mass diffusion between adjacent mesh elements, with the network learning the effective diffusion coefficients and interaction strengths from data. A stack of message passing layers, each of which iteratively updates the hidden states of nodes (and potentially edges) by aggregating information from neighbors. In each layer l , node states $\mathbf{h}_i^{(l)}$ are updated based on messages $\mathbf{m}_{ij}^{(l)}$ computed along edges (i, j) . A common scheme is to define:

$$\mathbf{m}_{ij}^{(l)} = \phi_{\text{edge}}^{(l)} \left(\mathbf{h}_i^{(l-1)}, \mathbf{h}_j^{(l-1)}, \mathbf{h}_{ij}^{(l-1)} \right) \quad (17)$$

We then aggregate messages from neighboring edges in some fashion, typically a sum or mean:

$$\mathbf{a}_i^{(l)} = \sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ij}^{(l)} \quad (18)$$

where $\mathcal{N}(i)$ is the set of neighbors of node i . Next, we update the node states:

$$\mathbf{h}_i^{(l)} = \phi_{\text{node}}^{(l)} \left(\mathbf{h}_i^{(l-1)}, \mathbf{a}_i^{(l)} \right) + \mathbf{h}_i^{(l-1)} \quad (19)$$

This process iterates for L layers, enabling information to propagate across the graph.

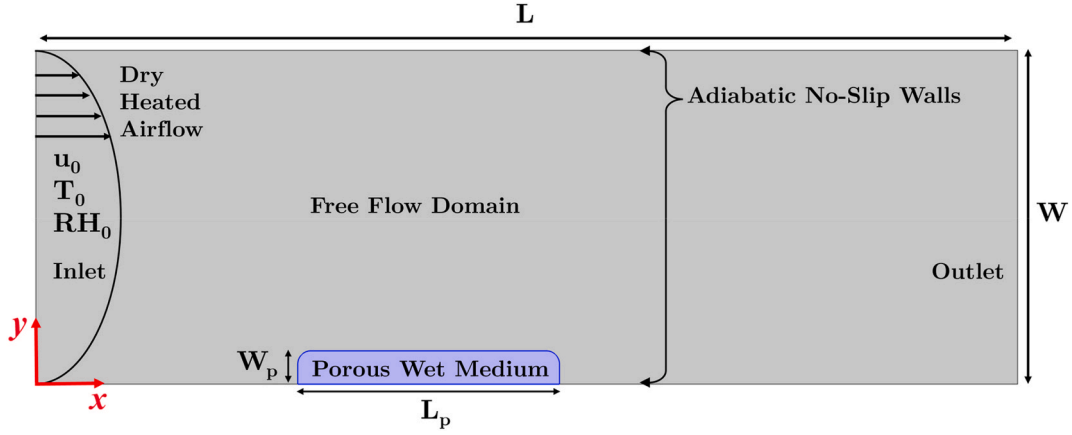


Fig. 1. Two-dimensional computational domain and boundary conditions for CFD simulation of convective potato drying. The rectangular channel ($L = 0.15$ m, $W = 0.05$ m) contains a centered potato slab ($L_p = 0.04$ m, W_p variable). Inlet boundary: prescribed velocity u_0 , temperature T_0 , and relative humidity RH_0 . Outlet: zero-gradient outflow. Top and bottom walls: no-slip velocity with adiabatic, impermeable conditions. The air-potato interface resolves conjugate heat and mass transfer without prescribed transfer coefficients. Simulations were performed in COMSOL Multiphysics v6.3.

c) **Decoder:** After the final message passing layer, we produce outputs that correspond to the target physical fields (e.g., predicted temperature and moisture at each node) by applying a decoder $\phi^{decoder}$:

$$\hat{\mathbf{y}}_i = \phi^{decoder}(\mathbf{h}_i^{(L)}) \quad (20)$$

Here, $\hat{\mathbf{y}}_i$ could be a multi-dimensional vector capturing the required physical variables for node i . Within the standard MeshGraphNet, this approach is used to learn surrogates for partial differential equation (PDE) solutions on a variety of meshes. However, for many real-world problems, particularly, those involving heterogeneous materials or regions with distinct physical properties standard message passing that treats all nodes and edges in a uniform manner may be insufficient. The problem defined here can exhibit complex multiphase and multi-region dynamics, particularly when the domain includes distinct material zones or phases (e.g., potato flesh, air pockets, and boundary regions). To address this complexity, our model incorporates a specialized masked message passing mechanism, which is detailed in the following section.

2.3. Masked message passing

In food drying, the physics at the air-food interface differs fundamentally from bulk transport within the food matrix or air domain. Standard neural networks struggle with these discontinuities, leading to nonphysical predictions like moisture appearing in air regions. The masked message passing extends the standard GNN framework to handle heterogeneous domains with distinct material properties (e.g., air regions and porous food material in drying applications). This approach enables selective message propagation and differentiated processing based on region type, particularly enhancing feature learning at interfaces where critical physics occurs. Unlike traditional message passing that treats all nodes uniformly, masked message passing identifies and processes different regions distinctly, enabling physics-informed feature learning. This is especially valuable in food drying applications, where heat and mass transfer mechanisms differ markedly between the air and porous media regions. In particular, the porous domain is governed by multiphase capillary and diffusive transport, whereas the air domain follows single-phase Navier–Stokes convection. As a result, a sharp discontinuity arises in the liquid saturation field at the interface, which standard message passing methods fail to capture. In the custom implementation, masked message passing is achieved through a modified processor layer that selectively processes nodes based on a region mask M . For each node i , $M_i = 1$ if node i belongs to the region of interest (e.g., porous media), and $M_i = 0$ otherwise. The masked message passing algorithm implements:

1. **Boundary Identification:** For regions specified by mask M , a boundary mask B is computed to include both region nodes and their direct neighbors:

$$B_i = M_i \vee \exists j : (M_j = 1 \wedge (i, j) \in E \vee (j, i) \in E) \quad (21)$$

2. **Selective Edge Processing:** Only edges where the destination node is in the boundary are processed:

$$E_{boundary} = \{(i, j) \in E | B_j = 1\} \quad (22)$$

3. **Focused Message Passing:** Messages are only propagated along edges in $E_{boundary}$:

$$\mathbf{a}_i^{(l)} = \sum_{j: (j, i) \in E_{boundary}} \mathbf{m}_{ij}^{(l)} \quad (23)$$

4. **Boundary Node Update:** Only nodes in the boundary have their embeddings updated:

$$\mathbf{h}_i^{(l)} = \begin{cases} \phi_{node}^{(l)}(\mathbf{h}_i^{(l-1)}, \mathbf{a}_i^{(l)}) + \mathbf{h}_i^{(l-1)} & \text{if } B_i = 1 \\ \mathbf{h}_i^{(l-1)} & \text{otherwise} \end{cases} \quad (24)$$

This focused computation significantly improves efficiency by concentrating computational resources on regions of interest and their boundaries, where the most relevant physical interactions occur. This modification ensures that the GNN's internal representation acknowledges regional boundaries and the mismatch of physical properties across them, a capability critical for capturing processes like moisture diffusion in heterogeneous media.

3. Methods

This study adopts a data-driven methodology to construct a surrogate model for food drying processes. In the following sections, we first describe the numerical simulation framework and GNN model configuration, then outline the experimental design strategy used to generate the training data, followed by procedures for organizing and processing the dataset, and conclude with a discussion of the training metrics and statistical methods employed for model validation.

3.1. Numerical simulation

All computational steps including mesh generation, implementation

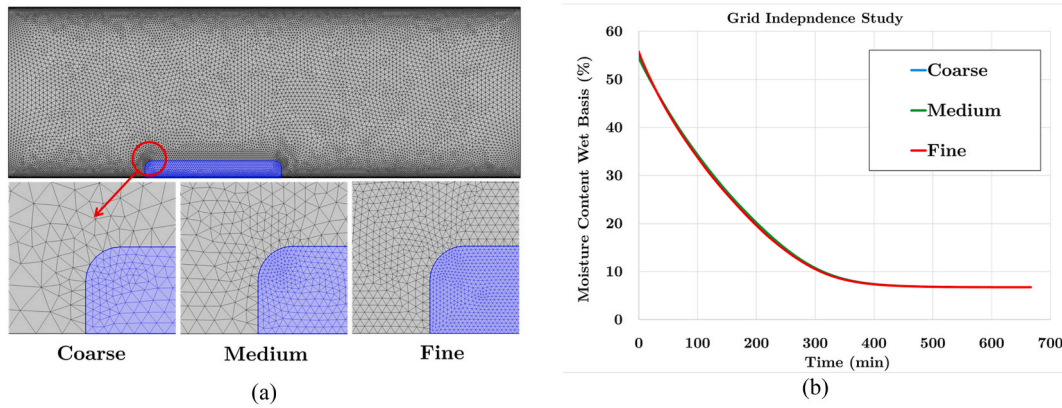


Fig. 2. Grid-independence assessment for the 2D conjugate CFD model. (a) Coarse, medium, and fine meshes evaluated (mesh statistics reported in Table 1). (b) Representative grid-independence comparison based on moisture-content predictions used to select the medium mesh for all subsequent simulations to balance computational cost and numerical accuracy.

Table 1

The overall statistics of grids used in the grid-independence study.

Grid	Coarse	Medium	Fine
Cell Type	Triangles	Triangles	Triangles
Cell #	5714	21376	83146
Node #	2971	10909	42014
Minimum Element Quality	0.4103	0.3549	0.3829
Average Element Quality	0.9157	0.9469	0.9541

of the governing partial differential equations, and numerical simulations were performed in COMSOL Multiphysics v6.3. A schematic of the two-dimensional computational domain and its principal boundary conditions is shown in Fig. 1. At the inlet, an ambient air free stream velocity u_0 is prescribed; a preliminary steady-state simulation is performed to achieve a fully developed velocity profile that is subsequently imposed in the main transient coupled heat and mass transfer calculations. Specified temperature T_0 and relative humidity RH_0 conditions are also applied at the inlet, while the outlet is treated as an outflow. Hereinafter, for simplicity we refer to the relative humidity as RH and liquid saturation as S_w . The top and bottom walls enforce no-slip flow

and are thermally insulated and impermeable, ensuring adiabatic conditions for heat transfer. A systematic grid-independence study was performed using three increasingly refined meshes (coarse, medium, and fine) to identify an optimal balance of accuracy and computational cost. Three triangular mesh resolutions were generated (see Fig. 2(a)) and their characteristics (number of nodes, cells, edges, and skewness values) are summarized in Table 1. Convergence was assessed by comparing predicted moisture distributions and flow fields among the three grids. Fig. 2(b) presents representative moisture content distributions for different mesh resolutions. Quantitative analysis demonstrates grid independence, with negligible variation under further mesh refinement. The Medium mesh exhibits a Root Mean Square Errors (RMSE) of 0.001 relative to the Fine mesh whereas the Coarse mesh shows larger deviations (RMSE = 0.0026). Based on these metrics, the Medium mesh was selected for subsequent simulations, as it provides results statistically equivalent to the Fine mesh at a substantially lower computational cost, making it well suited for large parametric studies.

3.2. Validation of numerical method

To validate the numerical framework employed in this study, we

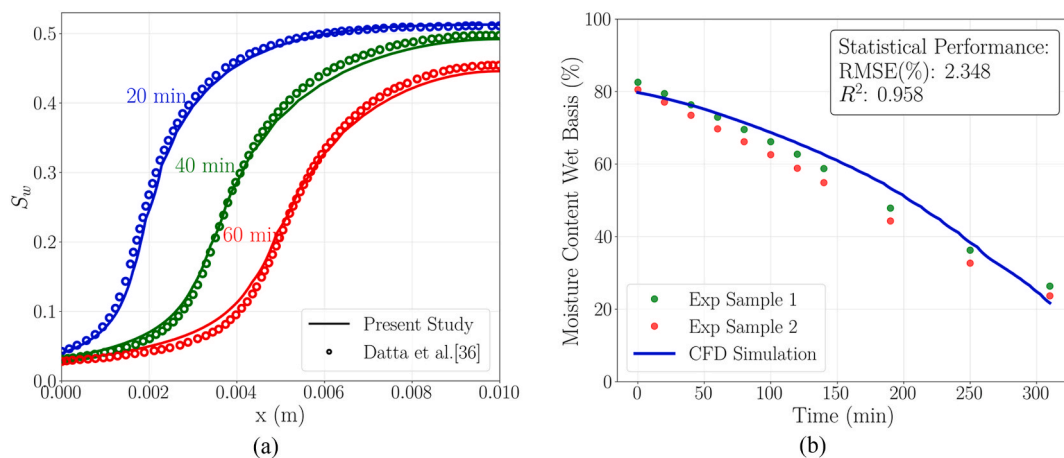


Fig. 3. Validation of the CFD simulation framework against numerical benchmarks and laboratory experiments. (a) Spatial profiles of liquid saturation (S_w) through the potato thickness at $t = 20, 40$, and 60 min, comparing present CFD simulation (solid lines) with reference data in (Datta, 2007b) (circles). Benchmark conditions: $T_0 = 177$ °C, $W_p = 1$ cm, prescribed quasi-steady transfer coefficients $h \approx 20$ W/m²K and $h_m \approx 0.01$ m/s at interface. Statistical analysis confirms no significant difference between the simulation and reference data ($p > 0.79$ for all timesteps, average Normalized Root Mean Square (NRMSE $\approx 2.0\%$). (b) Temporal evolution of wet-basis moisture content during convective drying of potato samples, comparing CFD prediction (blue line) with experimental measurements (Sample 1: green circles, thickness 6.2 mm; Sample 2: red triangles, thickness 6.1 mm). Experimental conditions: $T_0 = 42.2$ °C, $u_0 = 1.5$ m/s, $RH_0 = 13\%$, Statistical agreement: $R^2 = 0.958$, RMSE = 2.35% wet basis. Experimental apparatus details provided in Supplementary Fig. S3. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

performed a two-step validation. First, we compared our numerical simulations with a convective low-moisture potato drying case and evaluated the results against well-established reference data from (Datta, 2007b). This reference provides comprehensive analytical and numerical benchmarks for heat and mass transfer in porous food materials during drying processes. Second, we compared the simulation results with data obtained from our laboratory-scale hot-air drying experiments on potato samples in Section 3.3.

3.2.1. Validation against reference numerical and analytical benchmarks

The validation case considers a potato slab with dimensions optimized to approximate one-dimensional heat and mass transfer conditions while maintaining the full physics of the conjugate problem. The potato slab dimensions were set to $W_p = 1$ cm (width) and $L_p = 10$ cm (length), minimizing edge effects inherent in 2D geometries. The thermophysical properties of the potato, air, and water materials were kept identical to those reported in (Datta, 2007b) as well as the latent heat of evaporation $\lambda = 2.435 \times 10^6$ J/kg applied on the porous-air interface. A critical distinction between our approach and the reference study lies in the treatment of boundary conditions. While (Datta, 2007b) employed simplified 1D approach with prescribed constant heat and mass transfer coefficients ($h = 20$ W/m²K and $h_m = 0.01$ m/s), our simulation explicitly resolves the heated air flow over the potato surface. This conjugate heat and mass transfer approach captures the spatial and temporal variations in the transfer coefficients that arise from the developing flow field and evolving surface conditions. To achieve comparable average transfer coefficients to the reference case, we carefully adjusted the channel geometry (height $W = 3$ cm and length $L = 30$ cm) and the freestream velocity $u_0 = 1.5$ m/s through parametric studies. The moisture content relationship with RH (also known as water activity denoted as a_w) was determined using the sorption isotherm data from (Halder et al., 2007), which corresponds to the modified Ratti et al. correlation evaluated at a reference temperature of $T = 25$ °C. This equilibrium formulation is consistent with the property datasets established for multiphase porous media benchmarks (Halder et al., 2007; Purlis, 2019), as shown in supplementary Fig. S1.

3.2.2. Results

Fig. 3(a) presents the comparison between our numerical predictions and the reference data from (Datta, 2007b) for the spatial distribution of S_w at three different time instances: 20, 40, and 60 min. The S_w profiles yielded RMSE of 0.011, 0.009, and 0.010 at 20, 40, and 60 min, respectively. Independent two-sample t-tests (Virtanen et al., 2020) confirmed no statistically significant difference between the simulation and reference distributions ($p = 0.80, 0.92, 0.96$ for 20, 40, and 60 min, respectively; $\alpha = 0.05$). In this statistical analysis, the p -value represents the probability that the simulation and reference data share the same distribution, while α serves as the threshold for rejecting this hypothesis; since $p > \alpha$ in all cases, the simulation is considered statistically indistinguishable from the benchmark. The maximum absolute deviation observed was 0.027 (saturation fraction) at 20 min, coinciding with the sharp moisture gradient near the surface. The model reproduces the drying front progression from the surface toward the core, including the initial steep gradient and the subsequent flattening of the profile. Deviations between the numerical results and reference data are attributed to: (1) inherent 2D transport effects despite geometry optimization, versus the 1D assumption in the reference; (2) spatially and temporally varying transfer coefficients from our conjugate approach (see Supplementary Fig. S2), which initially fluctuate before stabilizing near $h \approx 20$ W/m²K and $h_m \approx 0.01$ m/s, compared to the constant coefficients in the reference model; and (3) explicit resolution of air flow dynamics including boundary layer effects. The high p -values and low error metrics validate the CFD framework's accuracy and confirm the numerical framework as a reliable foundation for generating training data for the Graph Neural Network surrogate model.

3.3. Validation using laboratory-scale hot-air drying experiments

3.3.1. Experimental setup and sample preparation

To validate the numerical framework against experimental observations, convective hot-air drying experiments were conducted using fresh potato (*Solanum tuberosum*) samples in a controlled laboratory wind tunnel. The experimental setup consisted of an open-loop drying chamber with a rectangular cross-section, equipped with temperature and airspeed sensors, an air blower, and an air heater, as shown in Fig. S3(a) in the supplementary section. Airflow was generated by a centrifugal air blower (Ebmpapst, Farmington, Connecticut, USA). The air velocity was precisely controlled using a voltage regulator to achieve the desired flow rate. The incoming air temperature was adjusted by a heater controlled by a digital dual NTC temperature controller (STC-3008, Walfont), with an accuracy of ± 1 °C and resolution of 0.1 °C. Fresh potatoes were purchased from a local grocery store in Champaign, IL, USA, and were kept in refrigerator at 4 ± 2 °C for 24 h before the experiment to equilibrate the temperature. The potatoes were washed using tap water and peeled using a stainless-steel knife. Samples were then cut into rectangular slab geometry using precision cutting equipment to achieve uniform dimensions. Three rectangular samples were prepared with the following dimensions: length of 10 mm, depth of 55 mm, and thicknesses of 6.2 mm (Sample 1) and 6.1 mm (Sample 2) as illustrated in Fig. S3(b). The third sample was retained as a backup. The samples were oriented such that the airflow was perpendicular to the longest side (55 mm dimension), effectively replicating the two-dimensional flow configuration used in the CFD simulations (Fig. S3(c) and S3(d)).

3.3.2. Experimental procedure

Drying experiments were conducted according to American Society of Agricultural and Biological Engineers standard -ASABE S448.1- (Chemists, 2000). The drying air temperature (T_0) was maintained at 42.2 °C, the ambient relative humidity (RH_0) was 13%, and the air velocity (u_0) was 1.5 m/s. All settings were adjusted well before placing the samples inside the dryer to ensure that fully hydrodynamic and thermally steady-state conditions were achieved. At specified time intervals, both potato samples were removed from the drying chamber and weighed using a high-precision analytical balance with an accuracy of ± 0.0001 g (Ohaus Corp., Switzerland). The weight measurements were recorded to track moisture loss over time. The drying process was continued until the sample weight change became negligible, indicating that equilibrium moisture content had been reached.

3.3.3. Moisture content determination

After completing the drying experiment, both samples were transferred to aluminum pans and placed in a vacuum oven (Thermo Scientific Precision™ Compact Oven) operating at 105 °C for 3 h to completely remove any remaining moisture and determine the solid content. This procedure follows the NFTA Method 2.2.2.5 standard (Ahn et al., 2014) for moisture content determination. The vacuum oven method is the standard reference method for determining solid mass, from which wet basis moisture content at each time point during the drying process is calculated. The wet basis moisture content M_{wb} (%) was calculated based on the ratio of water weight to the total sample weight (water plus solid mass):

$$M_{wb} = \frac{W(t) - W_{dry}}{W(t)} \times 100 \quad (25)$$

where $W(t)$ is the sample weight at time t , and W_{dry} is the completely dried solid weight determined from the vacuum oven.

3.3.4. CFD simulation configuration for experimental validation

To enable direct comparison between experimental measurements and CFD predictions, a simulation case was configured to match the

experimental conditions. The computational domain geometry was adjusted to represent the rectangular slab configuration with dimensions matching Sample 1 (10 mm × 55 mm × 6.2 mm). The simulation employed the following boundary conditions: inlet air temperature of 42.2 °C (315.35 K), ambient relative humidity of 13%, and free-stream velocity of 1.5 m/s. The initial moisture content of the fresh potato sample was set to 80% wet basis (4.0 kg_{water}/kg_{dry matter} on dry basis), consistent with typical values for fresh potato tissue.

The simulation was run for the same total duration as the experimental drying period, with results extracted at time intervals corresponding to the experimental measurement points. Temperature, relative humidity, and liquid saturation fields were computed throughout the drying process, with particular attention to the spatial distribution of moisture within the porous potato sample.

3.3.5. Comparison of experimental and simulation results

Fig. 3(b) compares experimental moisture content measurements with CFD simulation predictions for potato samples over a drying period of 310 min. Simulation results reproduced the experimental drying kinetics, with $R^2 = 0.958$ and RMSE = 2.35% (wet basis) between simulated and measured moisture content over the tested conditions. The drying profile exhibits a nonlinear decline, dropping from an initial moisture content of approximately 80% to a final value near 25%. The simulation reflects the falling rate period where internal diffusion becomes the rate-limiting mechanism, though the transition from the initial surface evaporation phase is smoother in the model than in the experimental points. Experimental variability is observable, with Sample 2 (red markers) drying slightly faster than Sample 1 (green markers). This variation likely stems from natural heterogeneity in potato tissue properties and minor differences in sample thickness (6.1 mm vs. 6.2 mm). The CFD simulation initially aligns with Sample 2 and overlaps Sample 1 between 40 and 60 min but subsequently overpredicts moisture content relative to both samples during the bulk of the falling rate period until approximately 270 min, after which it drops below Sample

2. These results suggest that the two-dimensional approximation employed in the CFD framework maintains sufficient predictive accuracy for thin rectangular samples where transport occurs primarily perpendicular to the airflow. The alignment between the physical model and experimental data, with an RMSE below 2.5%, supports use of the CFD dataset as training data for the GNN surrogate under the tested conditions.

3.4. Simulation setup and dataset generation

To explore the parameter space governing potato drying behavior, Latin Hypercube Sampling (LHS) was implemented to define simulation configurations (McKay et al., 1979). LHS is a statistical sampling technique that generates a near-random yet stratified distribution of input parameters, providing more uniform coverage of a multidimensional parameter space than simple random sampling. This method is particularly well suited for computationally intensive numerical simulations, as it enables efficient exploration of the input space with a limited number of simulation runs.

Three key parameters were identified as having the most significant influence on drying dynamics.

- Thickness of potato sample W_p : 3-8 mm, discretized to 1 mm increments
- Freestream velocity u_0 : 0.1-0.46 m/s, continuous range
- Ambient temperature T_0 : 40-80 °C, continuous range

The parameter ranges were selected based on typical industrial drying conditions for thin potato slices. For each parameter combination in the LHS design, a complete CFD simulation was executed using the numerical setup described in the previous section, capturing the coupled heat and mass transfer dynamics. To formalize the LHS procedure, let $\mathbf{X} = [X_1, X_2, X_3]$ represent the vector of parameters where $X_1 = W_p$, $X_2 = u_0$, and $X_3 = T_0$. For each parameter X_i , the range $[a_i, b_i]$ was divided

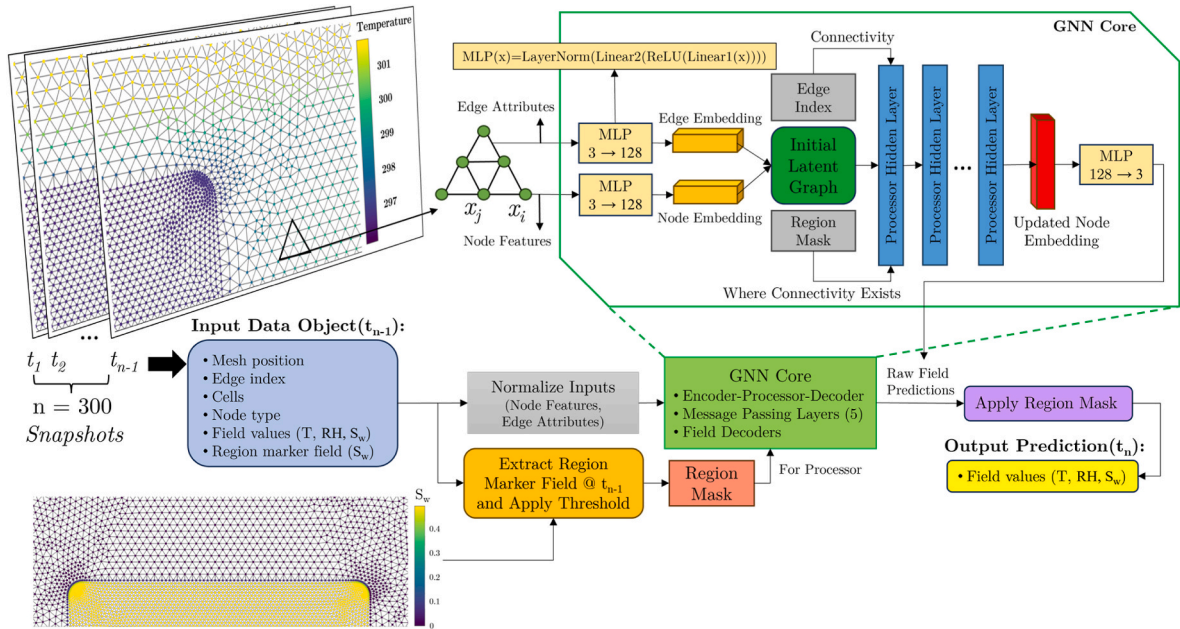


Fig. 4. Schematic of the region-aware Graph Neural Network (GNN) framework illustrating the end-to-end data flow from input state (t_{n-1}) to prediction (t_n). The pipeline begins with the Input Data Object, which aggregates mesh topology (positions, edge indices) and physical fields (Temperature T , Relative Humidity RH , Liquid Saturation S_w). Data processing branches into two parallel streams: (1) normalization of node and edge features, and (2) Region Mask generation, where the S_w field is thresholded (value > 0) to dynamically outline the porous food domain from the air. Inside the GNN Core, normalized features are projected into high-dimensional latent embeddings via independent node and edge Multi-Layer Perceptron (MLP) encoders. These embeddings enter the Processor, which executes sequential message passing; here, the Region Mask is utilized to restrict the propagation of S_w related information to valid nodes and edges, preventing non-physical leakage into the air domain. Finally, the Decoder projects the updated latent states back to physical field increments. The Region Mask is applied at the final time to the raw predictions to enforce strict boundary conditions before computing the final output state at t_n .

into $n = 40$ intervals. The sampling procedure ensured that for each dimension i , exactly one sample point fell within each interval:

$$P_{j,i} = \frac{\pi_j - U_{j,i}}{n}, j = 1, 2, \dots, n \quad (26)$$

where $P_{j,i}$ represents the probability for the j th sample in dimension i , π_j is a random permutation of integers $\{1, 2, \dots, n\}$, and $U_{j,i}$ is a uniform random number in $[0, 1)$. The final sample points were then computed as:

$$X_{j,i} = a_i + P_{j,i}(b_i - a_i) \quad (27)$$

This approach generated 40 parameter combinations spanning the experimental domain using SciPy's Latin Hypercube implementation (SciPy, 2025a) highlighted in Supplementary Fig. S4. Additionally, six validation points were strategically selected on the edges of the domain by generating 1000 random candidates and selecting those with maximum Mahalanobis distance (Mahalanobis, 1936) from the distribution center (Edge Cases). The Mahalanobis distance (SciPy, 2025b), which accounts for correlation between variables, was calculated as:

$$D_M(\mathbf{X}) = \sqrt{(\mathbf{X} - \mu)^T \Sigma^{-1} (\mathbf{X} - \mu)} \quad (28)$$

where $\mathbf{X} = [W_p, u_0, T_0]$ represents a parameter combination μ is the mean vector of the original LHS samples, and Σ^{-1} is the inverse covariance matrix of the original dataset.

3.5. Datasets

The dataset for training our GNN model consists of 40 simulation cases, which were divided according to a 70/15/15 split (%), resulting in 28 cases for training, 6 for validation, and 6 for testing. Each case contains both static and time-varying data organized as follows.

- **Static features:** These remain constant throughout the simulation and include:
 - a. mesh_pos: Node positions in 2D space with shape $[1, N_{node}, 2]$
 - b. cells: Triangular mesh elements with shape $[1, N_{cell}, 3]$
 - c. node_type: Integer labels identifying boundary and interior nodes with shape $[1, N_{node}, 1]$
- **Dynamic features:** These evolve over time and contain the physical fields:
 - a. temperature (T): Temperature ($^{\circ}\text{C}$) field with shape $[N_{\Delta t}, N_{node}, 1]$
 - b. humidity: RH (%) field with shape $[N_{\Delta t}, N_{node}, 1]$
 - c. [0-1] liq_sat (S_w): Liquid saturation field with shape $[N_{\Delta t}, N_{node}, 1]$

where N_{node} , N_{cell} , $N_{\Delta t}$ represent number of nodes, cells, and timesteps for each case, respectively. It is worth noting that in all simulations timestep $\Delta t = 100\text{s}$ and total physical time $t_{total} = 30,000\text{s}$ (500 min) resulting $N_{\Delta t} = 300$ for each case. The data processing pipeline involved parsing COMSOL mesh and field outputs converting them to PyTorch tensor format suitable for graph learning. Special attention was given to ensure consistent tensor shapes across different mesh sizes.

3.6. GNN parameters

Our GNN framework was implemented using PyTorch v2.5.1, with PyTorch Geometric (PyG v2.7) package providing specialized graph neural network primitives and data structures. Additional functionality was achieved through torch-scatter library for efficient scatter operations crucial for message passing implementations. Fig. 4 depicts the overall Graph Neural Network architecture used in this study, highlighting temporal data snapshots, particularly at t_{n-1} , the region-based masking logic that thresholds and applies material markers, the encoder-processor-decoder communication flow via successive message passing layers, and the sequential operations required to reconstruct the

mesh-based field variables at t_n .

3.6.1. Input features and normalization

The GNN model processes three principal physical fields that characterize the food drying process: Temperature, RH, and S_w . Each field possesses distinct characteristics that influence both their physical behavior and computational treatment within the network architecture. These fields are also interconnected and have reciprocal relationships with each other. TE and RH are implemented as global fields applied uniformly across all mesh nodes regardless of material region. S_w quantifies the moisture content within the porous food material and exhibits fundamentally different behavior from the ambient fields. It is implemented as a region-specific field meaning its values are only physically meaningful within the porous media domain. This field serves a dual purpose beyond representing a physical quantity, it also functions as the region marker field with a threshold value of 0 to delineate the boundary between porous media and air regions. The normalization strategy differs according to each field's physical characteristics. All fields undergo standardization to ensure numerical stability:

$$\hat{\mathbf{x}} = \frac{\mathbf{x} - \mu}{\sigma + \text{eps}} \quad (29)$$

where $\text{eps} = 10^{-8}$ prevents division by zero. We define mean $\mu \in \mathbb{R}^{d_x}$ and standard deviation $\sigma \in \mathbb{R}^{d_x}$ which are computed differently for each field. For temperature and RH, global normalization utilizes statistics computed across all mesh nodes. Conversely, S_w employs region-specific normalization, with statistics derived exclusively from nodes within the porous media region:

$$\mu_{S_w} = \frac{1}{|M|} \sum_{i \in M} x_i \quad (30)$$

$$\sigma_{S_w} = \sqrt{\frac{1}{|M|} \sum_{i \in M} (x_i - \mu_{S_w})^2} \quad (31)$$

where $M = \{i | M_i = 1\}$ represents the set of nodes within the porous media region. The model implements dynamic loss weighting that automatically adjusts the relative importance of each field during training based on their normalized scales. For field f , the effective weight is:

$$w_f^{\text{eff}} = w_f^{\text{static}} (1 - \alpha) + \frac{1}{\sigma_f + \text{eps}} \alpha \quad (32)$$

where $w_f^{\text{static}} = 1$ for all fields in the current configuration, and α is a warmup factor that transitions from 0.1 to 1.0 over the first 5 epochs. This mechanism ensures balanced learning across fields with different inherent scales and variabilities.

3.6.2. Loss function and optimization

The model's accuracy is measured using mean squared error (MSE) which quantifies the difference between predicted and actual values for each physical field (T , RH, S_w) in their physically relevant regions:

$$\mathcal{L} = \sum_{f \in \mathcal{F}} w_f \frac{1}{|M_f|} \sum_{i \in M_f} (\hat{y}_{f,i} - y_{f,i})^2 \quad (33)$$

where M_f is the subset of nodes where field f is evaluated. For region-specific fields like liquid saturation, this includes only nodes within the porous media region, while global fields (T and RH) are evaluated across all interior nodes, excluding boundaries where Dirichlet conditions are applied. This selective evaluation ensures the model focuses on learning the physics within domains where each field is physically meaningful, preventing incorrect penalization for undefined values (such as S_w in air regions). With the loss function defined, we now describe the network architecture that generates these predictions.

3.6.3. Network layers and latent dimension

The GNN architecture uses 128 internal variables at each mesh point to represent the physical state, a balance between capturing the complex physics and maintaining computational speed. The network processes information through 5 sequential layers, where each layer exchanges heat and mass transfer information between neighboring mesh elements. This depth allows the model to capture interactions up to 5 mesh elements away while preventing the loss of local detail that occurs with deeper networks. Each processing layer focuses computational effort on the porous food region and the air-food interface where the dominant processes occur evaporation, moisture diffusion, and heat transfer. The input and output components of the network use standard neural network building blocks (two-layer MLPs with ReLU activation) that transform between physical quantities and the network's internal representation. To enable training on standard computing hardware, the implementation uses memory-saving techniques (gradient checkpointing) that reduce requirements by 30% without affecting accuracy, allowing the model to process 16 drying scenarios simultaneously during training. This configuration was selected based on validation performance for the considered drying cases.

3.7. Training metrics and statistical analysis

3.7.1. Training configuration and convergence

The GNN model was trained for 75 epochs using the *AdamW* optimizer (Loshchilov and Hutter, 2019) with learning rate 1×10^{-4} and weight decay 5×10^{-5} . The learning rate was dynamically adjusted using *ReduceLROnPlateau* scheduling (Bengio et al., 2015), which halves the rate when validation loss plateaus for 20 epochs, to improve training stability. Training employed automatic mixed precision (AMP) to reduce memory usage while maintaining numerical stability, required for processing the high-dimensional mesh data. Early stopping monitored validation loss, terminating training after 30 epochs without improvement exceeding 1×10^{-4} , preventing overfitting. This adaptive approach resulted in nine distinct learning rate stages throughout training, ultimately reaching 3.9×10^{-7} by the final epochs (Supplementary Fig. S5(a)). Training showed distinct learning phases typically when teaching neural networks complex physical behaviors: rapid initial improvement, followed by oscillations as the model refines its understanding, then steady convergence to optimal performance. As shown in Fig. S5(b), the initial epochs (0-3) demonstrated rapid convergence with exponential loss reduction from approximately 10^2 to 1. This was followed by a period of oscillatory behavior (epochs 3-20) before entering steady convergence. The validation loss stabilized at approximately 0.53 after 74 epochs, with training and validation losses remained similar in later epochs as evidenced by the consistent parallelism between training and validation curves. It is worth noting that training follows a teacher forcing approach, where the model learns to predict the next timestep using ground truth data from the current timestep as input (Bengio et al., 2015; Pfaff et al., 2021). Throughout this study, 'ground truth' refers to the high-fidelity CFD simulation results from COMSOL Multiphysics, which serve as our reference data for training and validation. While not experimental measurements, these CFD solutions have been validated against well-established data in the literature (Section 3.2) and represent the physical behavior we aim to reproduce with the GNN model. This standard sequence modeling technique enables efficient learning of local physics at each data snapshot but may contribute to error accumulation in extended autonomous predictions when the model relies on its own outputs rather than ground truth inputs.

3.7.2. Field-specific performance

The architecture exhibited differentiated learning patterns across the three physical fields being modeled (Fig. S5(c and d)). Temperature implemented as a global field, exhibited the most dramatic improvement during training, with validation error decreasing by three orders of

Table 2

RMSE for the present GNN model across different rollout horizons.

Field	1-step	5-step	10-step	50-step	Full rollout
T (°C)	0.17	0.34	0.64	2.80	4.31
RH (%)	0.26	0.81	1.46	4.34	13.05
S_w [0–1]	0.0028	0.0029	0.0051	0.0163	0.0532

magnitude from initial values of 428.58 to final values of approximately 2.6×10^{-4} . The RH field similarly showed substantial improvement, though with more moderate gains, stabilizing around 1.6×10^{-2} in validation error. The S_w field, implemented as a region-specific quantity constrained to the porous media domain, presented the most challenging learning task. Its validation loss experienced substantial oscillations between epochs 4-10 before stabilizing around 4.9×10^{-2} by epoch 30. This behavior aligns with the physical complexity of moisture transport in porous media, which involves sharp gradients at the air-food interface and nonlinear coupling with temperature.

4. Results and discussion

4.1. Statistical evaluation

The trained GNN model was evaluated using a multi-horizon prediction framework to assess both short-term accuracy and long-term stability. The assessment incorporated rollout predictions at various time horizons (1, 5, 10, 50 steps, and full trajectory) to gauge the model's capacity for autonomous simulation. To quantify prediction accuracy while accounting for the differing scales of physical quantities, we employed Root Mean Square Error (RMSE). For each field f , the RMSE was computed as:

$$RMSE_f = \sqrt{\frac{1}{N_{node}} \sum_{i=1}^{N_{node}} (\hat{y}_i^f - y_i^f)^2} \quad (34)$$

where $\hat{y}_i^f$ represents the predicted value of field f at node i , y_i^f is the corresponding ground truth from CFD, and N_{node} is the number of nodes where the field is evaluated.

The results in Table 2 demonstrate that the model for single-step predictions exhibits RMSE values (≤ 0.0026) for RH , ($\leq 0.29\%$) for S_w , and ($\leq 0.17^\circ\text{C}$) for temperature. The model maintains reasonable accuracy for short rollouts up to 10 steps, with gradual degradation in performance as the rollout horizon increases. An interesting observation is the stability of water saturation predictions in the 1-step to 5-step range, where the RMSE remains nearly constant (0.0028 vs. 0.0029), suggesting the model has captured certain aspects of the moisture transport physics effectively for short time horizons. However, for longer rollouts, all fields show increasing error accumulation, with temperature exhibiting the most significant deterioration in full rollout predictions (with RMSE of 4.31°C). These results were achieved with an very limited dataset of only 28 training, 6 validation, and 6 test trajectories a stark contrast to the original MeshGraphNet implementation which utilized 1000/100/100 trajectories respectively (Pfaff et al., 2021). Furthermore, as previously mentioned, our model was trained using teacher forcing (also known as one-step prediction). Other researchers noted this approach can lead to error accumulation during extended autonomous predictions, as the model must rely on its own potentially imperfect outputs rather than ground truth inputs (Sanchez-Gonzalez et al., 2020). Even by providing a context of ground truth data from multiple previous timesteps during training as opposed to only one timestep, the error accumulation during long-term predictions was observed (Choi and Kumar, 2024). The use of purely data-driven architecture, i.e., MLP, for the encoder, processor, and decoder can be a potential source of error (Choi and Kumar, 2024). However, despite this inherent limitation, our model maintained

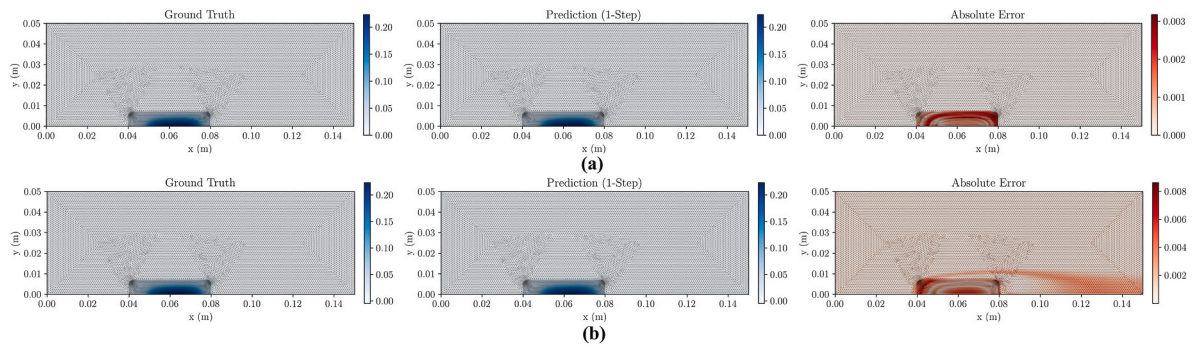


Fig. 5. Impact of masked message passing on liquid saturation (S_w) prediction. (a) Prediction with masking applied, correctly confining S_w to the porous media domain. (b) Prediction without masking, exhibiting non-physical “bleed-through” of liquid saturation into the air region due to unrestricted message passing. Absolute error maps $|S_{w, \text{predicted}} - S_{w, \text{CFD}}|$ show maximum error of 0.008 without masking, with systematic errors extending into the air domain, versus maximum error of 0.003 with masking (approximately threefold reduction), with errors localized at the material interface. Both configurations predict the porous medium interior with comparable accuracy, indicating that region-aware processing primarily improves handling of discontinuous fields at material boundaries. Test case conditions: $T_0 = 40^\circ\text{C}$, $u_0 = 0.25\text{ m/s}$, $W_p = 8\text{ mm}$.

respectable performance across different rollout horizons, with short-term predictions (1-10 steps) showing good accuracy before gradual degradation in longer autonomous simulations. Recent works have shown that models trained with teacher forcing can be made more robust for rollout prediction by incorporating noise during training or using scheduled sampling techniques (Brandstetter et al., 2023; Pfaff et al., 2021). Our implementation did not employ these techniques, indicating the inherent capability of the graph neural architecture to capture the underlying multi-physics of the drying process without specialized rollout training strategies. In practical digital twin applications, these long-term accuracy challenges could be potentially addressed through periodic data assimilation, where experimental measurements of surface moisture content and temperature are fed into the model at regular intervals to recalibrate predictions and prevent error accumulation over extended operational periods.

4.2. Impact of masked message passing on liquid saturation prediction

The region-aware architecture employs masked message passing to condition information propagation on material domain boundaries. To evaluate the efficacy of this approach, we compared S_w predictions with and without masking applied. Fig. 5 compares the predictions of S_w for an arbitrary case, showing the ground truth alongside predictions from both model variants and their corresponding absolute errors. In the ground truth, the S_w field is strictly confined to the porous media (potato slice), with zero values throughout the air domain, reflecting the physical reality that air does not contain liquid water. This sharp discontinuity at the material interface presents a significant challenge for graph neural networks, as unmasked message passing tends to propagate features across all connected nodes in the mesh. The model without masking exhibits significant bleed-through of S_w values into the air domain, with erroneous nonzero saturation extending several mesh elements away from the material interface. This phenomenon occurs because unmasked message passing allows water content information to propagate indiscriminately across the entire computational domain. Physically, this results in a violation of conservation principles and misrepresentation of the underlying physics. In contrast, the region-aware model confines S_w predictions to the porous media domain, maintaining sharp field discontinuities at the material interface. The absolute error plot for the region-aware model shows that errors are primarily localized at the food-air boundary, with maximum absolute error approximately three times lower than without masking (0.003 vs. 0.008). The interior of the porous media domain is predicted with good accuracy in both models, suggesting that the primary advantage of region-aware processing is in handling discontinuous fields at material interfaces. The reduction in maximum absolute error indicates that

conditioning message passing on region labels reduces interfacial bleed-through for discontinuous fields.

4.3. Fields contour predictions

To evaluate the spatial predictive capability of the model, we analyzed the temperature, RH , and S_w field contours at mid-drying timepoints (approximately $t_{\text{total}}/4$) for representative test cases. Fig. 6 presents comparisons between ground truth CFD simulations and GNN predictions for Case A (interior domain: $T_0 = 52^\circ\text{C}$, $u_0 = 0.24\text{ m/s}$, $W_p = 5\text{ mm}$) and Case B (edge domain: $T_0 = 80^\circ\text{C}$, $u_0 = 0.1\text{ m/s}$, $W_p = 8\text{ mm}$), focusing on the potato slice region and surrounding air interface. Contour predictions for two additional test cases are provided in Supplementary Fig. S6.

4.3.1. Temperature field predictions

The temperature field contours reveal agreement between the predicted and ground truth distributions across both cases. Case A and Case B demonstrate the model's ability to capture the temperature gradients emanating from the edges of the potato slice into its interior. The characteristic temperature depression in the potato slice is reproduced, reflecting the evaporative cooling effect during drying. For Case A, the absolute error plots indicate prediction discrepancies primarily concentrated as a ring band structure within the potato slice, with maximum errors of approximately 1.0°C , representing less than 0.3% error relative to the absolute temperature scale. For Case B, with its higher operating temperature ($\sim 80^\circ\text{C}$), the model exhibits error up to 1.25°C concentrated at the core section of the potato slice, corners, and partially extending into the air region. Despite this increase in error magnitude, the prediction accuracy remains within 2% relative error for a model operating at the edge of its training domain.

4.3.2. RH field predictions

The RH predictions demonstrate the model's capacity to capture the water vapor concentration gradients in the air domain surrounding the food product. For Case A, maximum absolute errors of approximately 3% correspond to less than 4% of the maximum RH value. Case B shows slightly higher errors up to 4%, still primarily concentrated at interfaces. Additional RH error distributions for interior and edge cases are presented in Supplementary Fig. S6.

4.3.3. S_w field predictions

The S_w contours provide the most demanding test of the model's capabilities, as this field is confined exclusively to the potato slice domain with sharp discontinuities at the material boundary. The GNN predictions capture both the magnitude and distribution of moisture

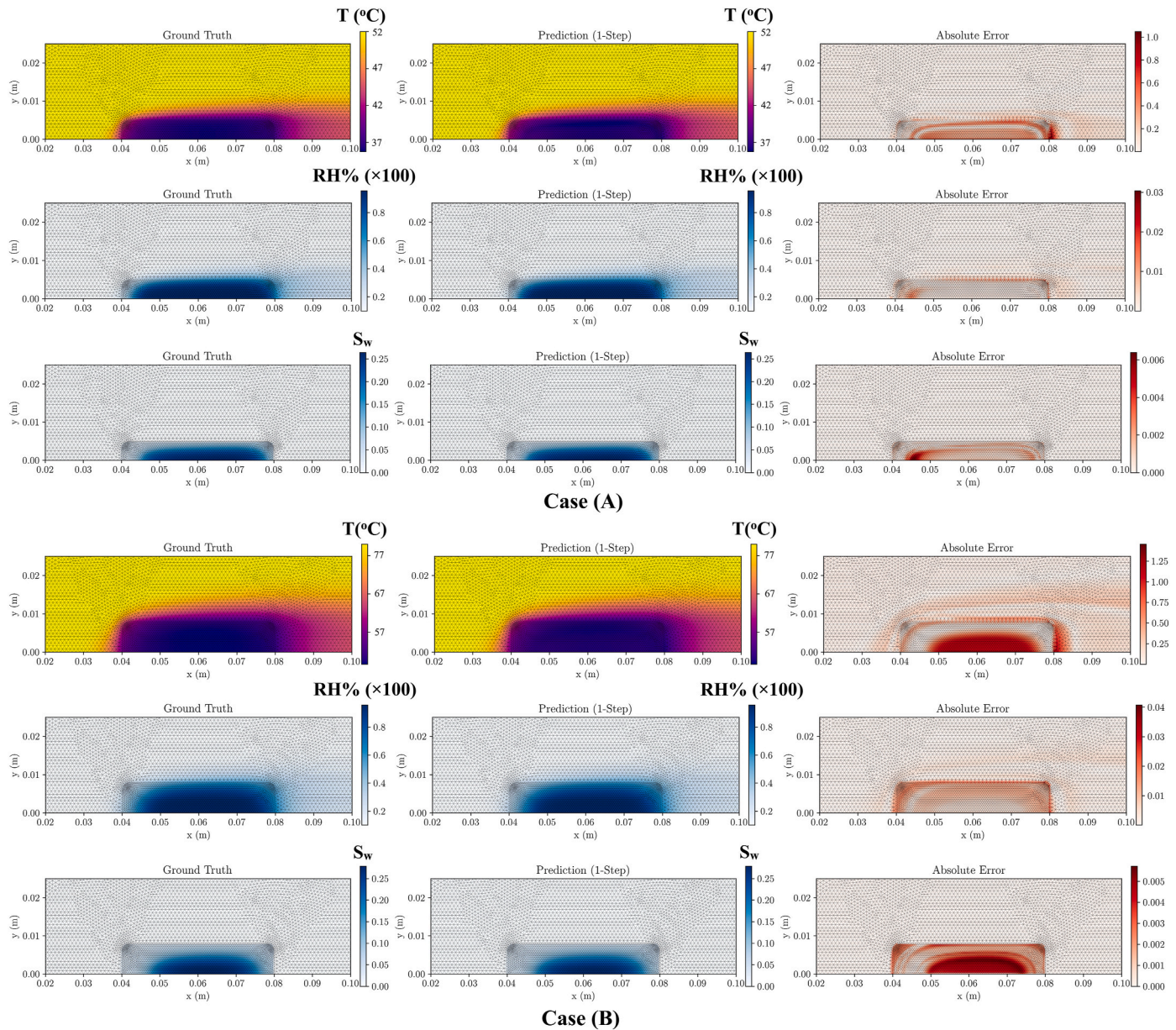


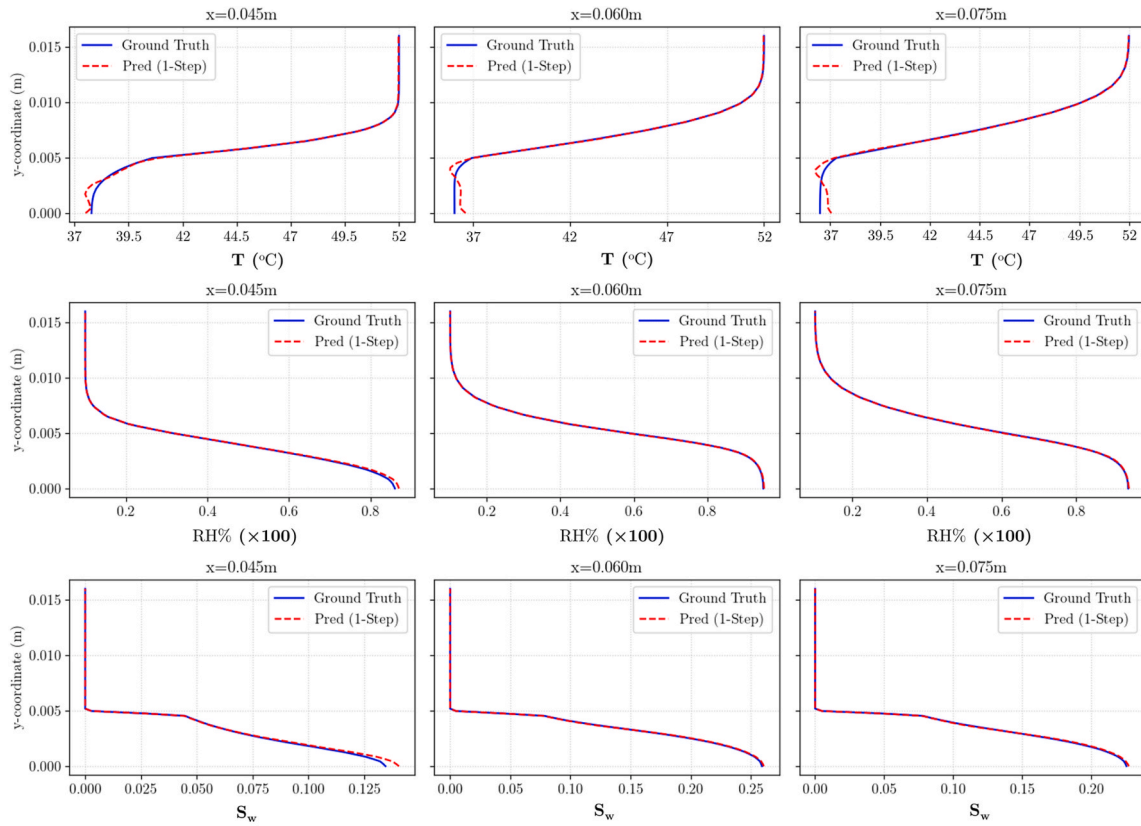
Fig. 6. Spatial field predictions from the GNN compared to ground truth CFD simulations at mid-drying timepoint for four test cases spanning interior and edge regions of the training parameter space. Case A (interior domain): parameter combinations within the LHS training distribution. Case B (edge domain): parameter combinations at maximum Mahalanobis distance (Mahalanobis, 1936) from training center, testing generalization capability. For each case: columns show temperature T ($^{\circ}\text{C}$), relative humidity RH (%), and liquid saturation S_w (0-1) fields, with corresponding absolute error maps. Case conditions A: $T_0 = 52$ $^{\circ}\text{C}$, $u_0 = 0.24$ m/s, $W_p = 5$ mm; B: $T_0 = 80$ $^{\circ}\text{C}$, $u_0 = 0.1$ m/s, $W_p = 8$ mm. Maximum absolute errors: $T \leq 1.25$ $^{\circ}\text{C}$ ($<2\%$ relative), $RH \leq 4\%$, $S_w \leq 0.006$.

within the potato slice for both cases. The error distributions reveal maximum discrepancies of 0.006 for Case A and 0.005 for Case B.

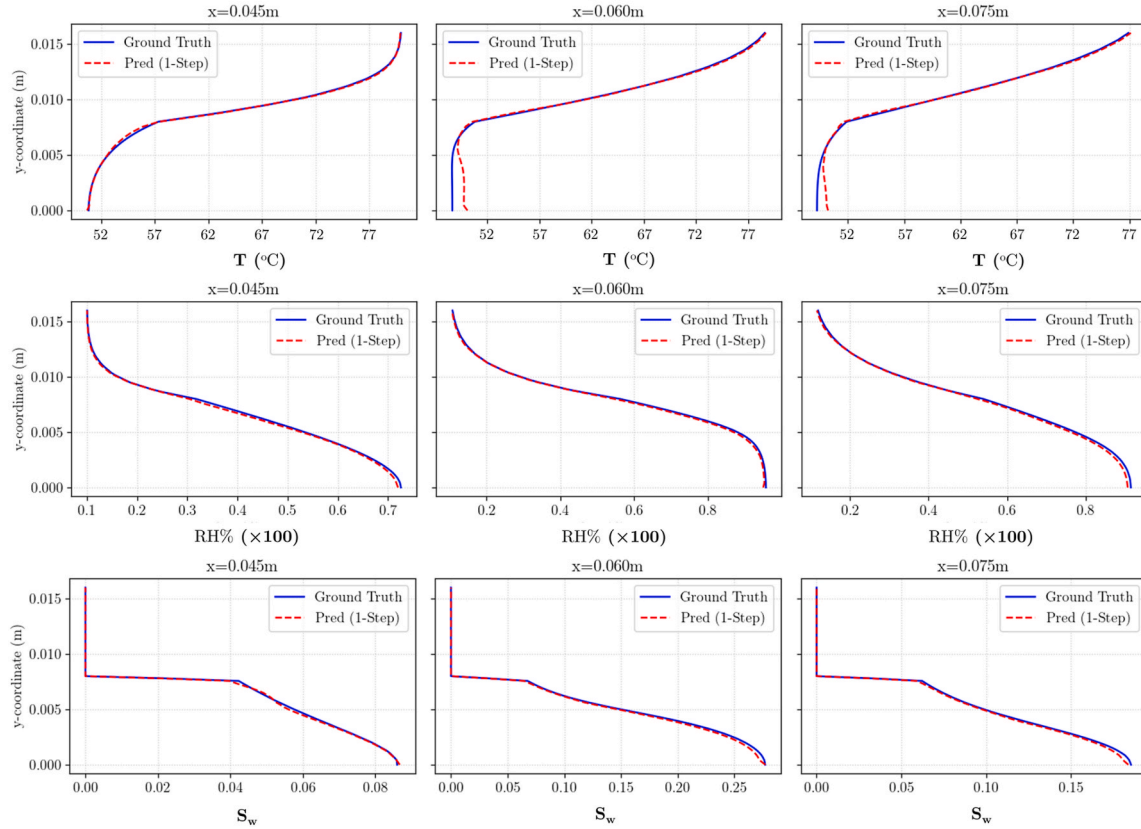
4.3.4. Physical significance and mesh adaptability

The spatial distribution of prediction errors across all three fields highlighted in Fig. 6 provides valuable insight into the model's physical learning. The consistent pattern of higher errors near material boundaries, particularly inside the potato slice adjacent to corners, reflects a fundamental challenge in multiphysics modeling: capturing the steep internal gradients that develop in response to the interfacial transport phenomena. These subsurface regions experience complex multidirectional heat and mass fluxes that create demanding computational conditions, especially where geometric features concentrate the physical effects. Despite these localized errors, the global error metrics remain low relative to the field ranges, and the predicted fields preserve the key

spatial patterns observed in the CFD reference solution. The model's ability to accommodate irregular meshes is evidenced by its consistent performance across both test cases, despite their different potato geometries and mesh structures. This mesh adaptability is crucial in food processing applications, where product shapes vary naturally and meshes must adapt to capture important features. In contrast to many reduced-order models that assume a fixed spatial discretization, the proposed GNN surrogate is evaluated on test cases with different mesh resolutions and connectivities than those used for training, without requiring mesh matching or field interpolation. It was shown that the model handled the multiscale nature of the potato drying process, capturing both the convective patterns in the air domain and the diffusive processes within the potato slice.



Case (A)



Case (B)

(caption on next page)

Fig. 7. Quantitative cross-sectional profiles comparing GNN predictions (red dashed lines) against ground truth CFD simulations (blue solid lines) along three vertical probe lines. Profiles extracted at front ($x = 0.045$ m), middle ($x = 0.060$ m), and aft ($x = 0.075$ m) positions relative to the potato slab for Case A (interior domain, top) and Case B (edge domain, bottom). Case A ($T_0 = 52$ °C, $u_0 = 0.24$ m/s, $W_p = 5$ mm): parameter combination within the LHS training distribution. Case B ($T_0 = 80$ °C, $u_0 = 0.1$ m/s, $W_p = 8$ mm): parameter combination at maximum Mahalanobis distance (Mahalanobis, 1936) from training center, testing generalization at parameter space boundary. For each case, rows show: (top) temperature T (°C) profiles exhibiting characteristic evaporative cooling within the potato slab ($y \approx 0$ – 0.005 m for Case A; $y \approx 0$ – 0.008 m for Case B) and thermal boundary layer development in the air domain; (middle) relative humidity RH (%) profiles capturing moisture accumulation near the potato surface and vapor transport into the airflow; (bottom) liquid saturation S_w (0–1) profiles demonstrating sharp discontinuity at the air–potato interface ($S_w = 0$ in air) and internal moisture gradients within the porous medium. The GNN accurately reproduces field distributions across all probe locations for both cases, with largest deviations occurring near the bottom potato surface where multidirectional heat and mass fluxes create steep gradients. Corresponding spatial field contours shown in Fig. 6. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

4.4. Profile analysis

To quantitatively assess the developed model's predictive capabilities along specific spatial dimensions, the field values along vertical probe lines at three distinct x -coordinates: near the front ($x=0.045$ m), at the center ($x=0.060$ m), and near aft ($x=0.075$ m) of the potato slice. These probe lines traverse both the potato slice and surrounding air regions, providing detailed cross-sectional views that complement the field contour analysis presented in the previous section.

4.4.1. Temperature profile analysis

The temperature probe line plots (Fig. 7) reveal good agreement between predicted and ground truth temperature distributions for both interior and edge cases. The model captures the characteristic S-shaped temperature profiles, with pronounced cooling in the potato slice region due to evaporative effects. For interior domain case (Cases A), temperature predictions maintain good alignment with ground truth across all three probe locations, with only minor deviations at the bottom half of the potato slice ($y \leq W_p/2$). For Case B ($T_0 = 80$ °C), the model performs well despite operating on the edge of its training domain. The probe analysis reveals that temperature prediction errors gradually increase toward the bottom of the potato slice ($y=0$), with similar error patterns observed at both the middle ($x=0.06$ m) and aft ($x=0.075$ m) of the sample. This corresponds with the plume-like or ring-band error structures visible in the contour error maps (Fig. 6), where maximum discrepancies occur not at the interface but within the interior regions of the potato slice. Beyond this narrow transition zone, temperature predictions regain high accuracy at the interface and free stream regions.

4.4.2. RH profile analysis

The RH probe plots (Fig. 7) demonstrate the model's ability to capture moisture transport phenomena in the vapor phase. All cases exhibit characteristic RH profiles with values increasing toward the potato slice bottom surface ($y = 0$) where the least amounts of heat and mass transfers occur. For both Case A and Case B, deviations from the CFD reference are small and are most evident near the bottom surface. This is consistent with the low one-step RH RMSE reported in Table 2. The model's RH predictions maintain fidelity across different operating conditions, from moderate to elevated temperatures.

4.4.3. S_w profile analysis

The liquid saturation probe plots (Fig. 7) provide the most stringent test of the model's capabilities, as they clearly delineate the porous media boundary where saturation abruptly drops to zero. The model demonstrates predictive performance for this discontinuous field, maintaining high accuracy across all cases despite the sharp material interface. Within the potato slice, liquid saturation generally decreases from the bottom toward the exposed surfaces, reflecting moisture migration during drying. The model captures these gradients, with prediction curves closely following ground truth for both cases.

4.4.4. Integrated physical insights

By combining field contour visualizations with quantitative probe line analysis, we gain comprehensive insights into the model's predictive

capabilities. The probe data confirms that the model has learned the coupled physics of heat and mass transfer during potato drying, with errors primarily concentrated at material's corners and bottom surface where sharp gradients and multidirectional heat and mass fluxes pose a challenge. The model's performance remains robust even on the edges of its training domain highlighted in Fig. 6, with Case B (highest temperature, edge of training domain) showing prediction accuracy comparable to the interior-domain Case A. The probe analysis (Fig. 7) also reveals the model's ability to handle the different physical behaviors across regions capturing both the diffusion-dominated transport within the potato slice and the convection-influenced profiles in the surrounding air.

4.5. Moisture reduction drying dynamics

4.5.1. Moisture ratio curves

The Moisture Ratio (MR) is a critical parameter in food drying analysis, representing the normalized moisture content of the product relative to its initial value. Mathematically, it is defined as:

$$MR = \frac{M - M_e}{M_0 - M_e} \quad (35)$$

where M is the instantaneous dry basis moisture content (kg of solid/kg of water) at time t , M_0 is the initial dry basis moisture content, and M_e is the equilibrium dry basis moisture content. This dimensionless parameter provides a standardized measure of drying progression, enabling direct comparison across different drying conditions and food materials. To evaluate the predictive capability of our model, we examined drying curves for two test cases: Case A (a parameter combination within the training domain) and Case B (a parameter combination on the edge of the training domain). For each case, we performed predictions with increasing levels of autonomy, from simple one-step predictions to complete autonomous rollouts, as shown in Fig. 8.

4.5.2. Within-domain predictions (Case A)

For Case A, the model demonstrates short-horizon predictive capability. The one-step evaluation yields $R^2 = 1.000$ and $RMSE = 0.003$ relative to the CFD reference solution. This accuracy persists through the 5-step rollout ($R^2 = 0.997$) and 10-step rollout ($R^2 = 0.989$), indicating that the model has learned the short-term physics of moisture transport within the porous media. As the autonomous prediction horizon increases to 50 steps, a modest divergence emerges between predicted and ground truth curves, particularly in the early drying stage (0–83 min). This divergence becomes more pronounced in the full rollout prediction, where the model consistently predicts a slower drying rate compared to the ground truth. Nevertheless, the model maintains the overall characteristic S-shaped drying curve and predicts the equilibrium moisture point, with a coefficient of determination ($R^2 = 0.784$).

4.5.3. Edge-domain predictions (Case B)

For Case B, which represents conditions on the edge of the training domain, the model exhibits similar performance for short-horizon predictions. Both one-step and 5-step evaluations yield $R^2 = 0.999$ and $RMSE = 0.01$ relative to the CFD reference solution. The 10-step rollout

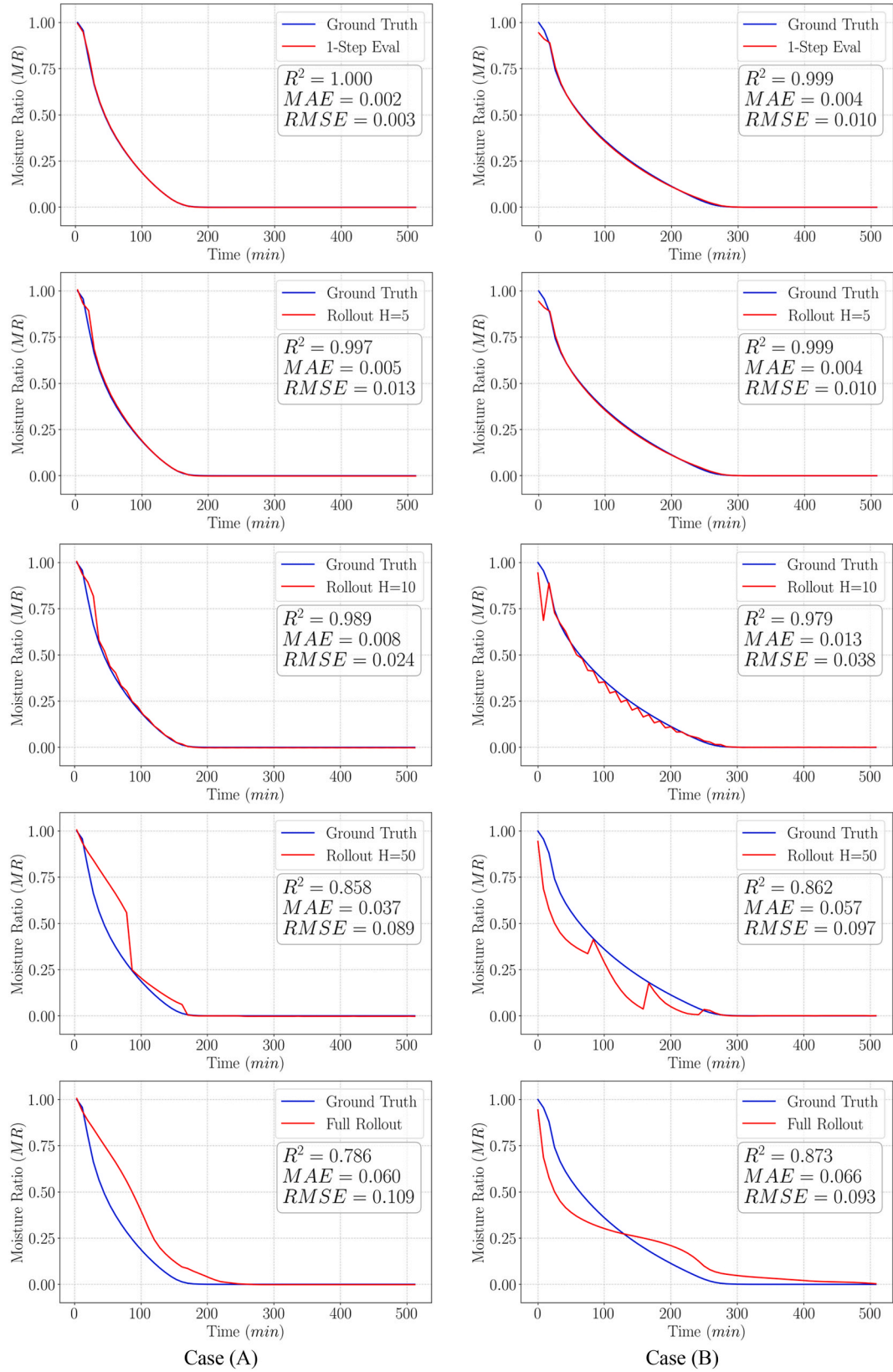


Fig. 8. Moisture Ratio (MR) drying curves comparing GNN predictions against ground truth CFD across increasing prediction horizons, evaluated for interior-domain (Case A) and edge-domain (Case B) test cases. **Case A** ($T_0 = 52^\circ\text{C}$, $u_0 = 0.24\text{ m/s}$, $W_p = 5\text{ mm}$): GNN achieves short-horizon accuracy (1-step: $R^2 = 1.000$; 10-step: $R^2 = 0.989$) and full autonomous rollout ($R^2 = 0.784$). (b) **Case B** ($T_0 = 80^\circ\text{C}$, $u_0 = 0.1\text{ m/s}$, $W_p = 8\text{ mm}$): Similar short-horizon performance (1-step: $R^2 = 0.999$; 10-step: $R^2 = 0.979$) despite operating at parameter space boundary; full rollout yields $R^2 = 0.873$. Both cases preserve characteristic S-shaped drying curves and correctly predict equilibrium moisture content. Divergence in extended rollouts stems from error accumulation inherent to autoregressive teacher-forcing training; accuracy for digital twin applications would be maintained through periodic data assimilation.

introduces minor oscillations in the prediction curve, with $R^2 = 0.979$. The 50-step prediction reveals more significant divergence and instabilities, particularly in the early and middle stages of drying. The full autonomous rollout demonstrates different behavior from Case A, with the model predicting an initially faster drying rate followed by a slower approach to equilibrium. Despite these timing differences, the model still achieves predictive performance with $R^2 = 0.873$.

4.5.4. Physical interpretation of model performance

The prediction patterns provide insight into how the model represents the drying dynamics. For short-horizon predictions, the model captures instantaneous drying-rate trends for cases both within the training domain and near its boundaries. This behavior is consistent with the model learning relationships that govern local moisture-transport dynamics in unsaturated porous media under the conditions represented in the training data. These results indicate that the surrogate can be coupled with periodic data assimilation to limit rollout drift in long-horizon predictions. The divergence in longer rollouts stems from error accumulation inherent to autoregressive prediction. Nevertheless, the preservation of the overall S-curve shape and accurate prediction of the equilibrium point demonstrates that the model has captured the fundamental physical behavior of the drying process. The contrasting behavior between Case A (slower predicted drying than ground truth) and Case B (initially faster, then slower drying) suggests that the model's sensitivity to operating conditions varies with position in the parameter space. This highlights the challenge of generalizing to conditions on the edges of the training domain, particularly for autonomous simulation over extended time horizons.

5. Conclusion

This study establishes a region-aware graph neural network surrogate framework for multiphysics porous media drying. By explicitly distinguishing food and environmental domains, the proposed framework incorporates domain-specific representations that reflect their distinct physical behaviors, leading to improved predictive accuracy across coupled fields while achieving two to three orders of magnitude computational speedup compared to traditional numerical simulations. The developed model achieved single-step RMSE of 0.17°C for temperature, 0.26% for RH , and 0.0028 for S_w , and maintained $R^2 > 0.97$ for 10-step rollout moisture ratio predictions. Several key findings emerged from the evaluation.

- Domain separation for Multiphysics problem: The proposed framework enforced separation between the food and hot-air domains, confining liquid saturation to the porous region and preventing nonphysical leakage of S_w values into the air domain. This design achieved a threefold reduction in interfacial absolute error (0.003 vs. 0.008). Such confinement was essential in the drying simulations, as S_w directly correlated with the moisture content.
- High predictive accuracy across physical domains: The model accurately captured the coupled multiphysics interactions among temperature, relative humidity (RH), and S_w fields in both the porous food domain and the surrounding air domain. It reproduced diffusion-dominated transport within the potato matrix and convection-driven behavior in the air region. Single-step predictions yielded RMSE values of 0.17°C for temperature, 0.26% for RH , and 0.0028 for S_w .
- Rollout performance and error accumulation: For moisture ratio curves, short-horizon rollouts maintained high fidelity ($R^2 > 0.97$ for 10-step predictions) for both the interior-domain case (Case A: $R^2 = 0.989$) and the edge-of-domain case (Case B: $R^2 = 0.979$), capturing the characteristic S-shaped drying behavior. Spatial field predictions showed progressive degradation with rollout length, with S_w exhibiting stability in the 1-5 step range (RMSE increasing only from 0.0028 to 0.0029).

The computational efficiency of the surrogate model, together with its predictive accuracy across distinct physical domains and short-horizon temporal stability, supports its use as a forward-model component in multiphysics industrial digital twin workflows for drying processes. Future work should evaluate integration with sensor networks in commercial dryers to enable online state estimation and predictive control. Combining sparse sensor measurements with the surrogate could provide estimates of internal moisture and temperature fields that are difficult to instrument directly and support adaptive control strategies targeting energy use and product quality. The graph-based formulation allows inference on geometries that differ from the training meshes, and region masks can be redefined to accommodate product-geometry variations. In addition, coupling the surrogate with reinforcement learning or other optimal-control methods is a potential direction for developing control policies across products and operating scenarios, with performance evaluated in terms of energy consumption and final product quality.

CRedit authorship contribution statement

Saeed Nazari: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Amir Malvandi:** Writing – review & editing, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this manuscript.

Acknowledgements

This study was financially supported by the Center for Advanced Research in Drying (CARD), a U.S. National Science Foundation Industry University Cooperative Research Center. CARD is located at Worcester Polytechnic Institute and the University of Illinois at Urbana-Champaign (co-site). We would also like to acknowledge Dr. Kamruzzaman and the Illinois Optical Sensing and Nanozyme Engineering Lab (IOSNEL) for providing access to the SMART Dryer, which enabled us to conduct the drying experiments. Additional support was provided by the Research Capacity Fund (HATCH), Project No. 7005794, from the U.S. Department of Agriculture, National Institute of Food and Agriculture.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jfoodeng.2026.113041>.

Data availability

Data will be made available on request.

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